Milestone 1: Rate–Distortion Benchmarking Suite

Classical vs. Learned Image Codecs

Omar Sameh Mahmoud
ID: 2501751
Dataset and Baseline Pipeline

David John Dawoud
ID: 2601189
Model and Algorithm Integration

Abdelrahman Refaat Saeed
ID: 2400004
Evaluation and Protocol Design

***Abstract*—This document presents the proposal and evaluation protocol for Milestone 1 of Project 30. The project develops a reproducible benchmarking pipeline comparing classical image codecs (JPEG, WebP, and AVIF) with pretrained learned image codecs through CompressAI. This milestone establishes dataset ingestion and preprocessing, integrates initial learned models, and specifies bitrate matching and evaluation using PSNR, MS-SSIM, LPIPS, and inference time.**

***Index Terms*—Image compression, rate–distortion, CompressAI, benchmarking, AVIF, learned codecs.**

I. Introduction and Objectives

Learned image compression (LIC) models require rigorous, reproducible comparisons against modern classical codecs. This project will build a unified benchmarking suite for classical codecs and pretrained learned codecs, including `bmshj2018-hyperprior` and `mbt2018-mean`, under a controlled evaluation protocol.

The objectives of Milestone 1 are:

- Assemble and preprocess the Kodak and CLIC validation datasets.
- Configure pretrained CompressAI models and verify inference.
- Define a reproducible bitrate-matching strategy and evaluation protocol.

Project repository: github.com/OmarSamehMahmoud/Rate-Distortion-Benchmarking-Suite

II. Team Roles and Responsibilities

- **Omar Sameh Mahmoud (2501751):** Dataset pipeline, classical codec baselines using Pillow/imageio, and data integrity checks.
- **David John Dawoud (2601189):** CompressAI setup, model instantiation, and forward-pass compatibility.
- **Abdelrahman Refaat Saeed (2400004):** Bitrate-matching design, metric implementation, and timing harness.

III. Dataset Specification

Two datasets will be used to provide a standard baseline and broader image coverage:

1) **Kodak:** 24 lossless images, typically 768×512 or 512×768. This dataset provides a sanity check and a point of comparison with prior work.
2) **CLIC validation set:** Professional-quality images. For learned codecs, image dimensions will be padded or cropped according to the selected model's requirements. Any cropping policy will be applied consistently across codecs to preserve a fair comparison.

IV. Protocol Specification

Classical codecs are typically controlled by a quality factor (QF), while learned codecs use model-specific quality levels or rate–distortion operating points. Therefore, QF will not be directly equated with a learned model's quality setting.

A. Bitrate-Matching Strategy

The protocol will use the following procedure:

1) Run each learned codec at a predefined set of CompressAI quality levels (initially levels 1–6) and record the resulting bits per pixel (bpp).
2) For each classical codec, search its quality-factor range for the setting whose bpp is closest to the learned codec's target bitrate.
3) Prefer matches within $\pm 1\%$ of the target bpp. If a codec cannot reach that tolerance because of discrete quality settings, report the achieved bitrate and the matching error rather than silently discarding the result.
4) Compare quality metrics at matched or nearest-achievable bitrate points, and clearly document any unmatched points.

B. Evaluation Metrics

- **Fidelity:** RGB PSNR and MS-SSIM.
- **Perceptual quality:** LPIPS with the AlexNet backbone.
- **Complexity:** Mean encode and decode time per image over 10 measured runs, excluding initial model loading. Hardware, software versions, and warm-up policy will be recorded.

V. Implementation Plan and Pipeline Architecture

The benchmarking suite will be implemented as a modular Python pipeline. Fig. 1 summarizes the processing stages and bitrate-matching controller.

A. Feasibility Test Results (Week 3)

A preliminary script, `benchmark_starter.py`, was executed to check the basic pipeline. The test loaded a Kodak image, confirmed dimensions divisible by 64, and ran a classical JPEG baseline and the pretrained `bmshj2018-hyperprior` model.

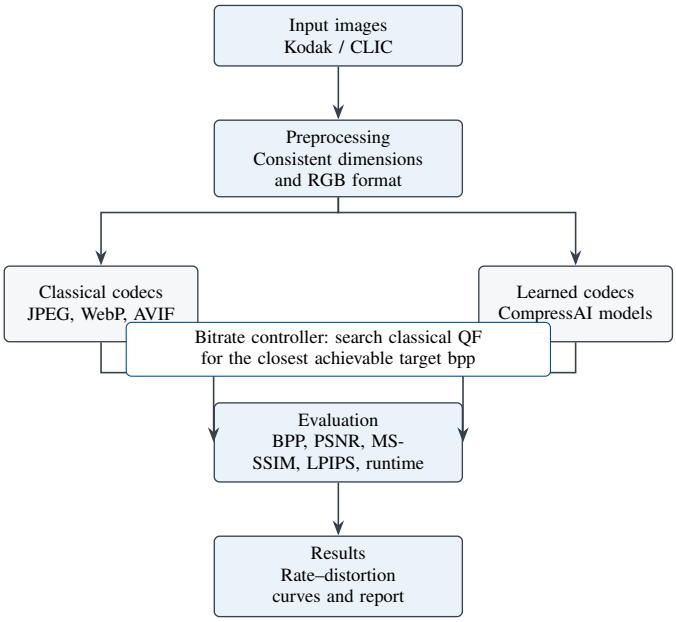


Fig. 1. Benchmarking pipeline. The bitrate controller matches classical codec settings to learned-codec operating points as closely as the available quality settings permit.

The recorded console output was:

```
================================================
Milestone 1: Feasibility Test & Pipeline Starter
================================================
[Preprocessing] Loaded image: 768x512
------------------------------------------------
[Classical JPEG] Quality: 75 | BPP: 1.8817
                 PSNR: 32.40 dB
[Learned Codec] bmshj2018-hyperprior (Q=3)
                 BPP: 0.5266 | PSNR: 28.68 dB
                 Encode time: 1.0882 s
================================================
[SUCCESS] Feasibility test passed.
================================================
```

These values are an initial feasibility check, not a bitrate-matched comparison: the two codecs produced substantially different bpp values. The full benchmark will compare codecs at matched or nearest-achievable bitrate points.

VI. Milestone 1 Deliverables

- ✓ Proposal and protocol specification.
- ✓ Dataset download and preprocessing starter script.
- ✓ Initial encode/decode feasibility test for a classical and a learned codec.
- ✓ Initial GitHub repository structure and `requirements.txt`.

References

References

[1] J. Balle, D. Minnen, S. Singh, S. Hwang, and N. Johnston, "Variational image compression with a scale hyperprior," in *International Conference on Learning Representations (ICLR)*, 2018. See also the CompressAI implementation: https://github.com/InterDigitalInc/CompressAI.

[2] Challenge on Learned Image Compression (CLIC), "CLIC validation datasets and tasks." [Online]. Available: https://clic2025.compression.cc/tasks/.

[3] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.